Communication-avoiding CG variants for the FESOM2 SSH solve

Derivation cross-check and discrepancy report

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Abstract

Four communication-avoiding solvers for the FESOM2 sea-surface-height (SSH) system — Chronopoulos–Gear CG (`cg2`), pipelined PCG (`pipecg`), PIPECG-OATI (`oati`) and the preconditioned Chebyshev/Stiefel iteration (`pcsi`) — were re-derived from first principles rather than transcribed, then cross-checked against three published listings and one existing Fortran implementation. Nine discrepancies were found. The most consequential is that the α recurrence shared by the whole Chronopoulos–Gear family is valid *only* for a symmetric preconditioner, a hypothesis stated in none of the sources; the FESOM MITgcm-class preconditioner is not symmetric, and the resulting error in the step length reaches 21.8 % on the production CORE2 matrix. This reproduces, and explains, an instability previously observed with `cg2`. A one-line remedy is given and verified.

1 Sources

[S]	S. Danilov, `solvers.F90` — `ssh_solve_cg2` and `ssh_solve_cg_pipelined` (FESOM2 working implementation).
[CV]	S. Cools, W. Vanroose, arXiv:1612.01395v3 — Alg. 4 (preconditioned Chronopoulos–Gear CG), Alg. 6 (preconditioned pipelined CG).
[P]	X. Huang *et al.*, *Geosci. Model Dev.* **9**, 4209–4225 (2016) — App. B2 (ChronGear), B3 (P-CSI), C (Lanczos).
[T]	M. Tiwari, S. Vadhiyar, *J. Parallel Distrib. Comput.* **163** (2022) 147–155 — Alg. 3–5 (PIPECG-OATI).
[GV]	G. Golub, R. Varga (1961); Y. Saad, *Iterative Methods*, Alg. 12.1 — Chebyshev semi-iteration.

Notation: A is the SSH stiffness matrix (symmetric to 1.4×10^{-13}, measured), M^{-1} the applied preconditioner, r the residual, $u = M^{-1}r$, $w = Au$, $\gamma = (r, u)$, $\delta = (w, u)$.

2 The principal finding: an unstated symmetry hypothesis

2.1 What the code builds

`fesom_ssh_preconditioner` constructs, per row i,

$$\mathrm{pr}[i,i] = \frac{1}{d_i}, \qquad \mathrm{pr}[i,j] = -\frac{1}{2}\,\frac{a_{ij}/d_i}{d_i + d_j} \quad (j \neq i), \qquad d_i \equiv A_{ii}. \tag{1}$$

*Report generated from the M10 track of the FESOM2 Kokkos port. Source of record: `docs/plans/20260805-m10-ssh-derivations.md`, branch `m10-ssh-solvers`.

The transpose entry carries d_j in the leading division, hence

$$\frac{\mathrm{pr}[i,j]}{\mathrm{pr}[j,i]} = \frac{d_j}{d_i} \quad\Longrightarrow\quad M^{-1} \neq (M^{-1})^\top \text{ wherever } d_i \neq d_j. \tag{2}$$

On an unstructured ocean mesh the diagonal varies with cell area and depth, so this is a structural property, not a small perturbation. Measured on the CORE2 matrix:

operator	$\max\lvert X_{ij} - X_{ji}\rvert$	$\max\lvert X_{ij}\rvert$	**defect ratio**
M^{-1} (`pr_values`)	8.561×10^{-8}	1.342×10^{-7}	**0.638**
A (control)	2.150×10^{-5}	1.510×10^{8}	1.42×10^{-13}

2.2 Why the baseline PCG is unaffected

Baseline PCG forms $\alpha_i = \gamma_i/(p_i, Ap_i)$ with an *explicit* dot product every iteration. Nothing is recurred, so no orthogonality identity is invoked, and the method converges normally.

2.3 Why every Chronopoulos–Gear variant is affected

The family replaces that explicit quantity by the recurrence

$$\sigma_i = \delta_i - \beta_i^2\sigma_{i-1}, \qquad \alpha_i = \gamma_i/\sigma_i, \qquad \sigma_i \equiv (p_i, Ap_i). \tag{3}$$

With $p_i = u_i + \beta_i p_{i-1}$ and A symmetric,

$$(p_i, Ap_i) = \delta_i + 2\beta_i(u_i, s_{i-1}) + \beta_i^2\sigma_{i-1}, \qquad s_{i-1} = Ap_{i-1}, \tag{4}$$

so (3) holds *if and only if*

$$(u_i, s_{i-1}) = -\beta_i\sigma_{i-1}. \tag{5}$$

Using $s_{i-1} = (r_{i-1} - r_i)/\alpha_{i-1}$,

$$(u_i, s_{i-1}) = \frac{(M^{-1}r_i,\, r_{i-1}) - \gamma_i}{\alpha_{i-1}}, \qquad (M^{-1}r_i, r_{i-1}) = (r_i,\, M^{-\top}r_{i-1}). \tag{6}$$

That last term vanishes by the standard PCG residual M-orthogonality *only when* $M^{-\top} = M^{-1}$; then $(u_i, s_{i-1}) = -\gamma_i/\alpha_{i-1} = -\beta_i\sigma_{i-1}$, which is (5). With a non-symmetric M^{-1} the term does not vanish, (5) fails, and σ_i drifts from the true (p_i, Ap_i) cumulatively, because σ is a recurrence. A wrong σ gives a wrong α, which is no longer the line-search minimiser: the method stops being CG.

2.4 Measured consequence

A diagnostic replay carries both the true (p_i, Ap_i) and the recurrence (3) through a reference PCG solve on a dumped CORE2 system ($n_{\mathrm{nz}} = 870\,146$), first with the preconditioner as built and then with the symmetrised form of §2.5:

iteration	as-built: rel. σ drift	as-built: $\lvert(u_i, r_{i-1})\rvert/\gamma$	symmetrised: drift
2	1.98×10^{-2}	1.28×10^{-2}	5.4×10^{-14}
8	8.20×10^{-2}	8.11×10^{-2}	8.3×10^{-14}
20	1.53×10^{-1}	1.66×10^{-1}	5.4×10^{-14}
40	1.77×10^{-1}	2.00×10^{-1}	3.8×10^{-14}
60	$\mathbf{2.18 \times 10^{-1}}$	$\mathbf{2.41 \times 10^{-1}}$	1.5×10^{-14}
worst	**21.8** %	**24.1** %	1.17×10^{-13}

The quantity required to vanish instead reaches 24 % of γ, and the step length is wrong by up to 22 % — at the production tolerance, on the production matrix. Running `cg2` in the model with the literal preconditioner fires the auto-fallback guard on **18 of 20 solves**, the residual growing to 2.9×10^4 against a tolerance of ≈ 4. With the symmetrised preconditioner: **zero fallbacks**, 123.35 iterations per solve. Independently, the α sequence departs from reference PCG by 2.017 % at iteration 2, matching the measured σ drift of 1.977 % term for term.

2.5 Remedy

Factor the preconditioner as $M^{-1} = D^{-1}C$ with $D = \mathrm{diag}(A)$, $C_{ii} = 1$ and $C_{ij} = -\frac{1}{2}a_{ij}/(d_i + d_j)$. Since $a_{ij} = a_{ji}$, the matrix C *is* symmetric — the entire asymmetry sits in the left factor D^{-1}. Define

$$\boxed{\widetilde{M}^{-1} := D^{-1/2}\, C\, D^{-1/2}} \qquad \text{i.e.} \qquad \widetilde{\mathrm{pr}}[i,j] = \mathrm{pr}[i,j]\sqrt{d_i/d_j}, \quad \widetilde{\mathrm{pr}}[i,i] = \mathrm{pr}[i,i]. \tag{7}$$

Two properties make this the right object:

1. $\widetilde{M}^{-1}$ is symmetric by construction.
2. $\widetilde{M}^{-1}$ is *similar* to M^{-1}, since $D^{1/2}(D^{-1}C)D^{-1/2} = D^{-1/2}CD^{-1/2}$. Similar matrices share their spectrum, so $\widetilde{M}^{-1}$ is exactly the same quality of approximate inverse. (Verified: $\lambda_{\max}(M^{-1})$ and $\lambda_{\max}(\widetilde{M}^{-1})$ agree to ten significant figures.)

Cost: one multiplication per stored entry, applied once at setup; the sparsity pattern and all halo machinery are unchanged.

Caveat. The similarity is a statement about M^{-1}, not about the preconditioned operator: $\widetilde{M}^{-1}A$ is not similar to $M^{-1}A$, and its extreme eigenvalue shifts by 0.23 % on a variable-diagonal test problem. Same convergence class, not the identical operator.

3 Discrepancy report

Severity: **S1** — wrong results if transcribed literally; **S2** — misleading or not executable as written; **S3** — cosmetic.

id	**source**	**issue**	**sev.**
T-1	[CV] Alg. 4; [S]; [T]	The α recurrence $\sigma_i = \delta_i - \beta_i^2\sigma_{i-1}$ is stated without its hypothesis. It is valid only for a symmetric M^{-1} (§2). The sources assume an "SPD preconditioner" in prose; the FESOM MITgcm-class `pr_values` is not symmetric (defect ratio 0.638). Measured consequence: α wrong by up to 21.8 % on CORE2, and `cg2` diverges on 18 of 20 solves.	**S1**
T-2	[P] App. B2	Line 4 computes σ_k using β_k, which is only defined on line 5: the listing is not executable in the order printed. Harmless in effect ($\sigma_0 = 0$ makes the undefined β_1 multiply zero on the first pass), but the reader must reorder from $k \geq 2$. Likewise $\rho_0 = 1$ gives $\beta_1 = \rho_1$ rather than 0, harmless only because $s_0 = p_0 = 0$.	**S2**
T-3	[T] Alg. 4, l. 11	The γ_{i+1} recurrence folds (r_i, q_i) and (s_i, u_i) into a single $2\alpha_i(\cdot)$ term. That identification requires $(r_i, M^{-1}w_i) = (M^{-1}r_i, w_i)$, i.e. a symmetric M^{-1} — the same unstated hypothesis as T-1.	**S1**

id	source	issue	sev.
T-4	[T] Alg. 4, l. 12	$\delta_{i+1} = \lambda_8 - \alpha_i(\lambda_1+\beta_i\lambda_2) - \alpha_i(\lambda_1+\beta_i\lambda_2) + \dots$ — the identical product is printed twice instead of $2\alpha_i(\cdot)$. Arithmetically equivalent, but it conceals that the two terms arise from *different* inner products which coincide only under a symmetric M^{-1}.	**S3**
T-5	[P] App. C	The Lanczos start vector is printed as $q_1 = r_0/(r_0^\top s_0)$: normalisation by the inner product rather than by its **square root**. The missing root corrupts α_1 and plants a spurious small eigenvalue in T. Measured on a 1024-unknown fixture: $\theta_{\min}$ returns 2.49×10^{-6} instead of 5.65×10^{-3} — **2270× too small**, and it does *not* wash out with m (identical error at $m = 30$ and $m = 120$). Feeding that ν to P-CSI inflates the assumed condition number by ∼2000×, so the Chebyshev polynomial is optimised for a far too wide interval and the method converges much more slowly than it should — silently, since it still converges.	**S1**
T-6	[P] App. B3, l. 1	The ω recurrence coefficient is typeset as a fraction that can be read as either $\frac{1}{4}\alpha^2$ or $1/(4\alpha^2)$. *Not a defect in the paper* — a transcription hazard. **Resolved to** $1/(4\alpha^2)$ three ways: by derivation (§4), by consistency with the paper's own seed $\omega_0 = 2/\gamma$, and against [GV]. Recorded so it need not be re-litigated.	**S1** *if misread*
T-7	[T] §2.2	The method is designed and benchmarked with **Jacobi** (diagonal) preconditioning, where M^{-1} requires no halo exchange. Not a typo, but a portability caveat: with a sparse M^{-1} carrying A's stencil, every preconditioner application consumes a halo ring, and the $n \to g \to h \to e \to f$ chain becomes four chained applications — roughly doubling the exchanged halo volume.	**S2**
T-8	[S] both routines	`MPI_Iallreduce` is followed immediately by `MPI_Wait` (`ssh_solve_cg2`: 37–38, 61–62, 99–100). A non-blocking collective with a zero-length overlap window is a blocking reduction *plus* the non-blocking surcharge, measured at $+8\,\mu$s on `openmpi/4.1.2` and $+1.6\,\mu$s on `openmpi/4.1.5-nvhpc`.	**S3** *perf*
T-9	[S] both routines	Hard-coded `Do iter=1,200` with no handling of iteration-limit exhaustion: a non-converged solve exits silently carrying whatever X holds.	**S2**

Nothing in [CV] Alg. 4 or Alg. 6 was found to be transcriptionally wrong; both listings reproduce the derivations line for line. The substantive issues are the unstated symmetry hypothesis (T-1, T-3) and the P-CSI Lanczos normalisation (T-5).

4 Resolution of the P-CSI ω recurrence (T-6)

Let the preconditioned operator have spectrum in $[\nu, \mu]$ and set $\theta = \frac{1}{2}(\mu + \nu)$, $\delta_{\mathrm{h}} = \frac{1}{2}(\mu - \nu)$, $\sigma_1 = \theta/\delta_{\mathrm{h}}$. The standard three-term form [GV] is $\rho_0 = 1/\sigma_1$, $\rho_k = 1/(2\sigma_1 - \rho_{k-1})$ and $\Delta x_k = \rho_k\rho_{k-1}\Delta x_{k-1} + (2\rho_k/\delta_{\mathrm{h}})M^{-1}r_k$. With the paper's variables $\alpha = 2/(\mu - \nu) = 1/\delta_{\mathrm{h}}$ and $\gamma = \theta$, put $\omega_k := 2\rho_k/\delta_{\mathrm{h}}$. Then

$$\omega_k = \frac{2}{\delta_{\mathrm{h}}(2\sigma_1 - \rho_{k-1})} = \frac{2}{2\theta - \delta_{\mathrm{h}}\rho_{k-1}}, \qquad \rho_{k-1} = \tfrac{1}{2}\delta_{\mathrm{h}}\omega_{k-1}, \tag{8}$$

so that

$$\omega_k = \frac{1}{\theta - \frac{1}{4}\delta_{\mathrm{h}}^2\,\omega_{k-1}} = \boxed{\frac{1}{\gamma - \dfrac{\omega_{k-1}}{4\alpha^2}}} \tag{9}$$

since $\delta_{\mathrm{h}} = 1/\alpha$. The reading $\frac{1}{4}\alpha^2$ is therefore incorrect. Two independent confirmations: the paper's own seed $\omega_0 = 2/\gamma$ follows from $\rho_0 = 1/\sigma_1$ only under this reading; and a direct numerical test shows the derived coefficient meets the Golub–Varga convergence bound at $k = 10, 20, 40$ while the misread one does not. Note also that the initial step uses $\Delta x_0 = \gamma^{-1}M^{-1}r_0$, *not* ω_0; a reader assuming otherwise is off by exactly a factor of two on the first step.

5 Verification

All statements above are reproducible from the M10 branch:

- symmetry defect and σ drift on real dumped matrices: `tools/fesom_ssh_lab -sym-check`, `-sigma-drift`;
- algorithm equivalence, Chebyshev rate, Lanczos normalisation (T-5) and the ω resolution (T-6): `tests/test_ssh_solvers.cpp` (18 assertions, host-only, runs in under a second);
- non-blocking-collective progression (T-8): `tools/iallreduce_probe.c`, run on both production MPI stacks.

Full derivations for all four solvers, the cross-source tables and the measurement provenance are in `docs/plans/20260805-m10-ssh-derivations.md`; the gate and measurement ledger is `docs/SSH_SOLVERS_M10.md`.